St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Endocrine and reproductive| Lecture 20

Development of the reproductive system

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Objectives

SOM.MKII.BPM2.1.ER.1.ANAT.1101	Describe the time-line for development of the genital system.
SOM.MKII.BPM2.1.ER.1.ANAT.1079	Describe the development of the gonads and the differentiation into testes and ovaries.
SOM.MKII.BPM2.1.ER.1.ANAT.1093	Discuss the level at which the gonads develop and its relationship to referred pain.
SOM.MKII.BPM2.1.ER.1.ANAT.1080	Describe the descent of the testes into the scrotum and the clinical significance of failure to descend.
SOM.MKII.BPM2.1.ER.1.ANAT.1083	Describe the development of the male reproductive ducts and associated glands.
SOM.MKII.BPM2.1.ER.1.ANAT.1084	Describe the development of the female reproductive ducts in relation to the paramesonephric ducts, uterovaginal primordium and urogenital sinus.
SOM.MKII.BPM2.1.ER.1.ANAT.1087	Describe the formation and differentiation of the external genitalia of males and females from the urogenital folds and labioscrotal swelling.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1464	Describe the development, location, blood supply, innervation and venous drainage of the suprarenal glands.

Recommended reading and resources

T.V.N. (Vid) Persaud, MD, PhD, DSc, FRCPath (Lond.), FAAA; Mark G. Torchia, MSc, PhD

The Developing Human, 12th ed. Chapter 12, pg. 225-264

Images:

Gary C. Schoenwolf, PhD; Steven B. Bleyl, MD, PhD; Philip R. Brauer, PhD; Philippa H. Francis-West, PhD

Larsen's Human Embryology, 6th ed. Chapter 16, pg. 386-418

Other resources:

Urogenital System. In: Sadler TW. eds. *Sadler15*. Lippincott Williams & Wilkins, a Wolters Kluwer business; 2024. Accessed July 18, 2025. https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?bookid=3221&sectionid=253323054

The Big picture

Sperm → Zygote
Secondary oocyte → Zygote
Zygote → Morula → Blastocyst
Blastocyst → Outer Cell Mass → Trophoblast
Trophoblast → Syncytiotrophoblast
Trophoblast → Cytotrophoblast
Blastocyst → Inner Cell Mass → Embryoblast
Embryoblast → Hypoblast → Umbilical vesicle
Embryoblast → Epiblast

Epiblast → Endoderm, Mesoderm, Ectoderm

Endoderm

- Epithelial lining of Gut and associated glands (liver, pancreas, gall bladder)
- Epithelial lining of the respiratory organs (trachea, bronchus, lungs), tonsils, middle ear and pharyngotympanic tube.
- Parathyroid and thyroid glands
- Epithelial lining of bladder, most of urethra and lower vaginal canal

Mesoderm

Lateral

- Heart
- Spleen
- Adrenals
- Parietal / Somatic Layer
- Visceral / Splanchnic Layer

Paraxial

- Sclerotome - bone
- Myotome - muscle
- Dermatome - skin

Intermediate

- Urogenital organs and ducts

Ectoderm

Ectoderm →
- Epidermis, nails, sweat glands, lens, anterior pituitary
- Sensory cells of the ear and olfactory epithelium, lower anal canal

Ectoderm → **Neuroectoderm** → Neural Tube, Neural Crest Cells

Neural Tube

- CNS
- Retina
- Pineal body
- Posterior pituitary

Neural Crest Cells

- PNS
- Melanocytes
- Glial & Schwann cells
- Cranial nerves
- Odontoblasts
- Connective tissue & bones of face & skull
- Cells of the adrenal medulla
- Bulbar and truncal ridges of the heart

Special note

Some slides have a green background, this will not be covered in the live lecture but are put so you have the information available

Some key points

- Most of what we know is based on animal studies with confirmation on human fetal material
- Anterior and posterior are **not synonymous** with ventral and dorsal in embryonic development
- Development of urinary and genital systems are closely related
- Development of the gonads depend on migration and invasion of primordial germ cells from the umbilical vesicle
- The pattern of sex chromosomes determines the path of development towards male or female phenotype
- The development itself is driven not only by genes on the sex chromosomes but also by hormones and other factors which are encoded on autosomes
- The SRY (**S**ex determining **R**egion of the **Y** Chromosome) gene is the switch, if absent or defective female development occurs
- Direct Sertoli cell to primordial germ cell communication is necessary for development of testes
- Genital system form from the intermediate mesoderm
- The first phases of development are identical on macro- and microscopic levels
  - Ambisexual or bipotential phase (previously referred to as indifferent)
- Ambisexual/ambiguous/bipotential phase ends at week 7
- Development of the female ductal system (particularly the vagina) is not fully understood

The journey of development

- Fertilization to gastrulation - Some primordial cell differentiation happens - **Weeks 1-3**
- Bilateral dorsal aorta develop on either side of the developing notochord – **Week 3**

<u>Developmental processes related to reproductive organs – Weeks 3 to 12</u>

Everything happens synchronously with many simultaneous events

- Formation of cloacal folds and genital tubercle – **Week 3**
- Formation of intermediate mesodermal urogenital ridge on either side of the dorsal aorta
- Formation and degeneration of pronephros
- Mesonephros and mesonephric duct develop
- Paramesonephric ducts develop
- Primordial germ cells start to migrate from the umbilical vesicle to the urogenital ridge

Week 4

- Primordial germ cells arrive at the gonadal ridge
- Fetal suprarenal cortex forms

Week 5

- Development of the genital swellings
- Formation of the primitive sex cords
- Gonadal development begins

Week 5 & 6

The journey of development cont.

- Ambisexual/bipotential phase ends
- Gonads start to show sex differentiation
- Neural crest cells migrate into developing suprarenal cortex

Week 7

- Fusion of caudal paramesonephric ducts
- Leydig cells (in males) start producing testosterone

Week 8

- Testes descend into inguinal canal
- External genital organs complete differentiation

Week 12

- Uterus and vaginal canal complete formation **- Week 20**
- Female gonads complete development of primordial follicles
- Male gonads descend into the scrotum

Weeks 28 to 38

- ♂ Seminiferous ducts acquire a lumen and spermatogenesis commences
- ♀ Primordial follicles begin maturation to secondary follicles every ±28 days

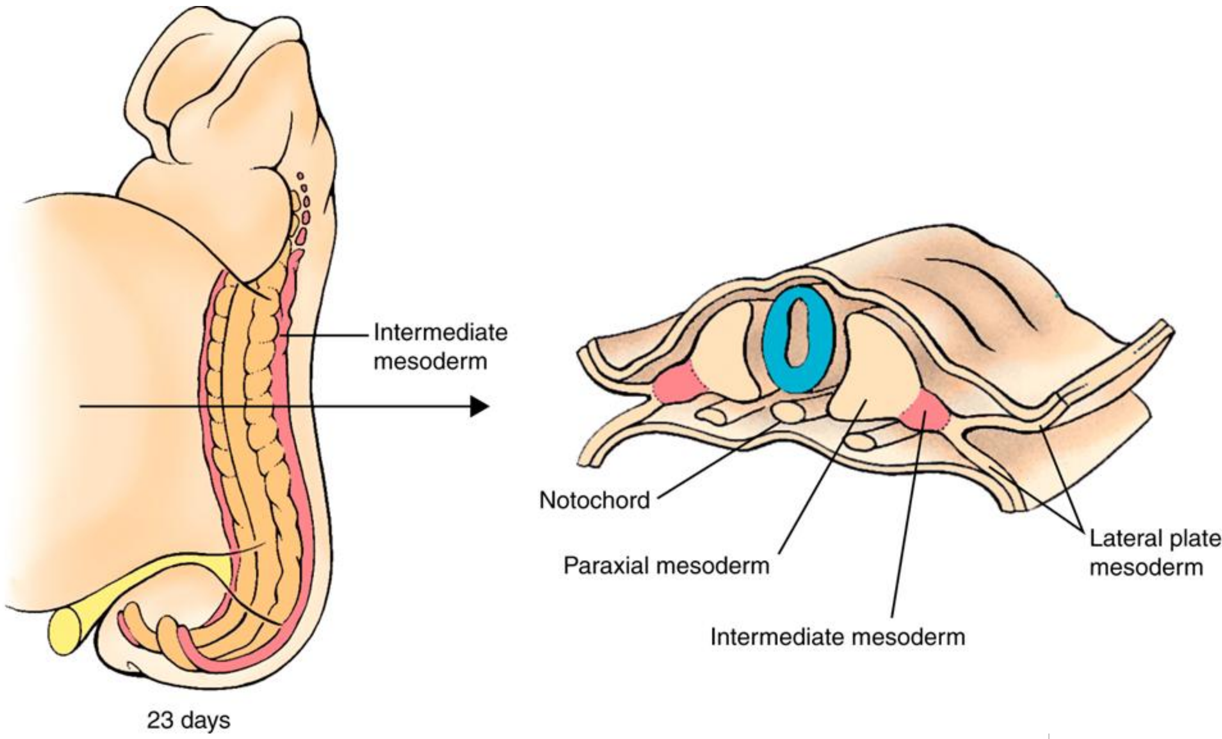
Puberty

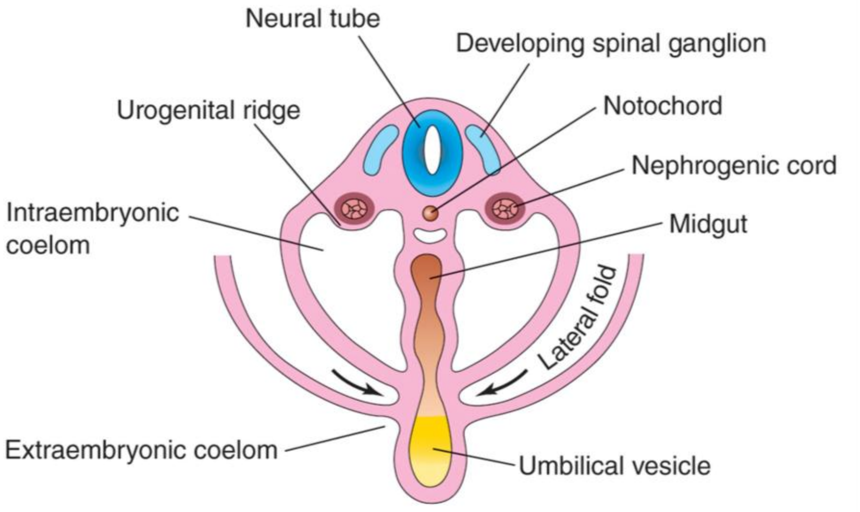
Lecture outline

- General concepts of development of the genitourinary system
- Ambiguous/bipotential phase of development
  - Gonads
  - Development of the suprarenal glands
  - Ducts (internal organs)
  - External genital organs
- Development when Y-Chromosome is present
  - Testes
  - Male genital ducts and accessory male reproductive glands
  - Male external genital organs
  - Formation of the inguinal canals and testicular descent
- Development when Y-Chromosome is Absent
  - Ovaries
  - Uterus
  - Vaginal canal
  - Female external genital organs
  - Final positioning of the ovaries and formation of the female ligaments

General concepts for development of the urogenital system

- Development of the urinary and reproductive system are linked
- Develops from
  - **Intermediate mesoderm**
    - Forms a ridge just lateral to the developing dorsal aorta = **urogenital ridge**
    - Deeper portion forms the **nephrogenic cord**
  - The most caudal end of the primitive gut tube
    - Expands to form **the cloaca**
  - Splanchnic mesoderm surrounding the hindgut
    - Gives rise to muscle and connective tissue of the bladder





Ambiguous/bipotential phase of development

- Gonads
- Development of the suprarenal glands
- Ducts (internal organs)
- External genital organs

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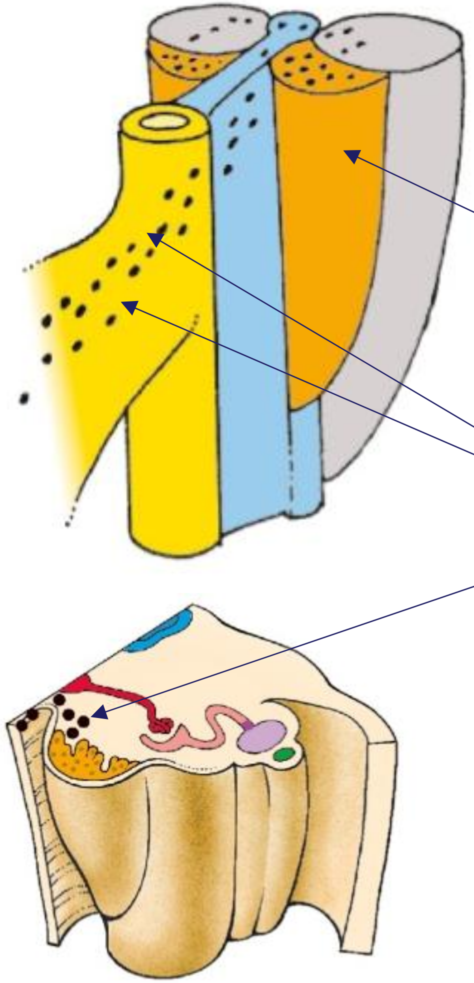
Week 3-6 - Ambiguous/bipotential phase of development

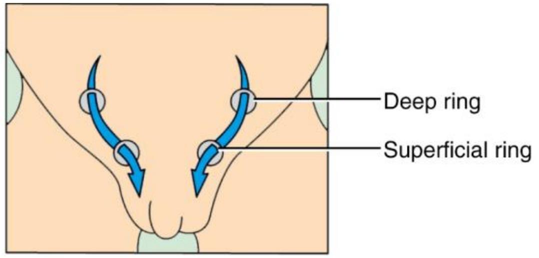
Gonads

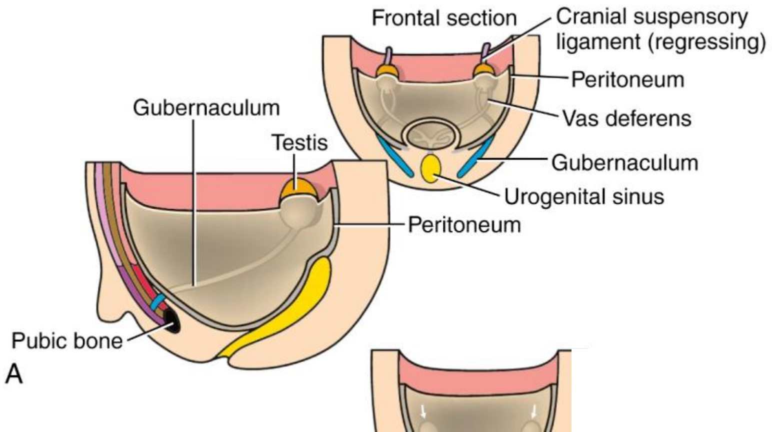


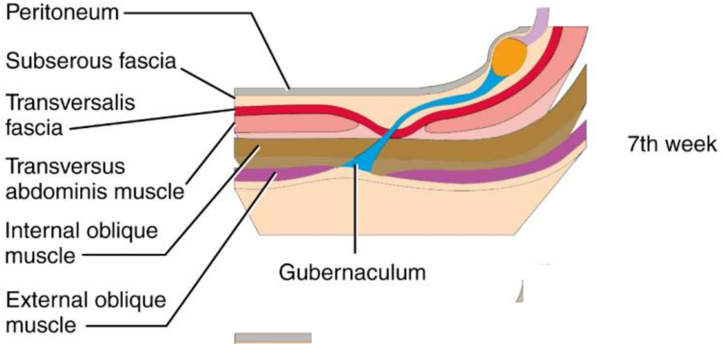
- **Primordial germ cells** - Epiblast cells migrate through the primitive streak and settles in the umbilical vesicle (Week 3)
- **Urogenital ridges** - Folding of the embryo pushes the mesenchyme of the intermediate mesoderm away from the somites causing bilateral bulges (week 4)
  - Deeper portions form the nephrogenic cords
  - Overlying epithelium proliferates and thickens forming the genital ridges
- Primordial germ cells (black dots) migrate into the genital ridges, corresponding to the **T10 vertebral level**, via the dorsal mesentery (week 4) – starts proliferating immediately
- **Primitive sex cords (medullary cords)** - Genital ridge epithelium proliferates and penetrate the mesenchyme (week 4)
  - Helps to guide and align the migrating germ cells

Ligaments of the developing gonads

- The developing mesonephric gonadal system lies in the **retroperitoneal fascia**
- The developing ductal system is also located retroperitoneally
- **Cranial suspensory ligament** attaches the superior pole and the diaphragm
- **Gubernaculum** is attached on the inferior pole of the developing gonad attaching it to the fascia between the developing external and internal oblique muscles in the region of the developing labioscrotal folds
  - Passes posteriorly to the mesonephric and paramesonephric ducts
- Present in both male and female embryos







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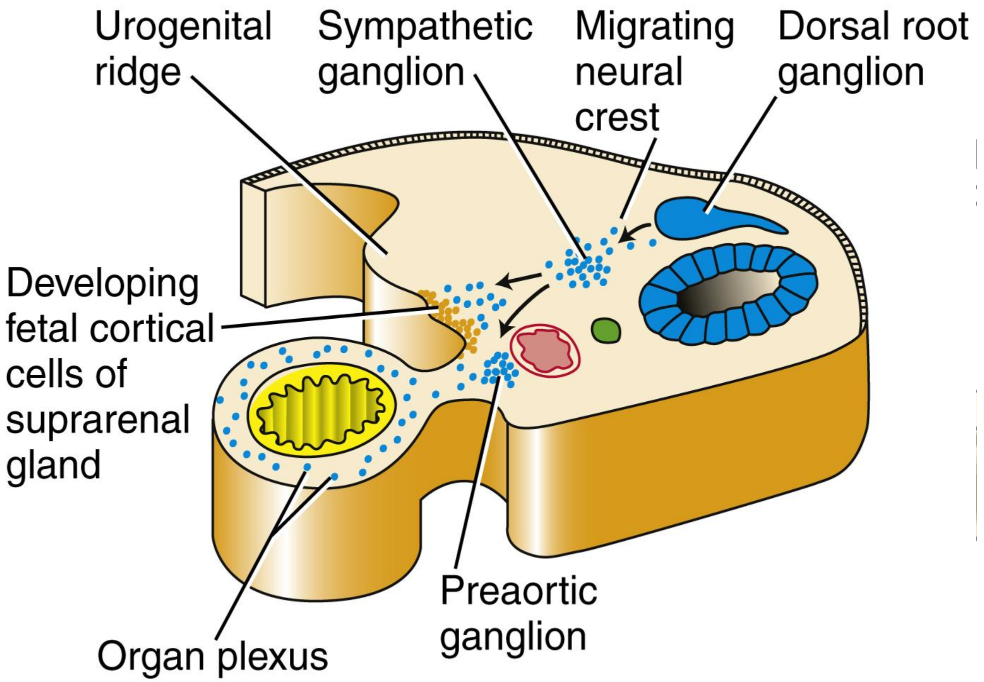
Development of the suprarenal glands

- **Coelomic epithelium cells** medial to the developing indifferent gonad proliferate and migrate into the underlying mesoderm ventral to the mesonephros **(Week 5)**
  - Differentiate into large acidophilic cells = **fetal suprarenal cortex**
  - A second set of cells develop surrounding the fetal cortex cells = **definitive cortex**
- These cells exhibit features of **steroid producing cells**
- **Neural crest cells** migrate into the developing fetal cortex and form the medulla **(week 7)**



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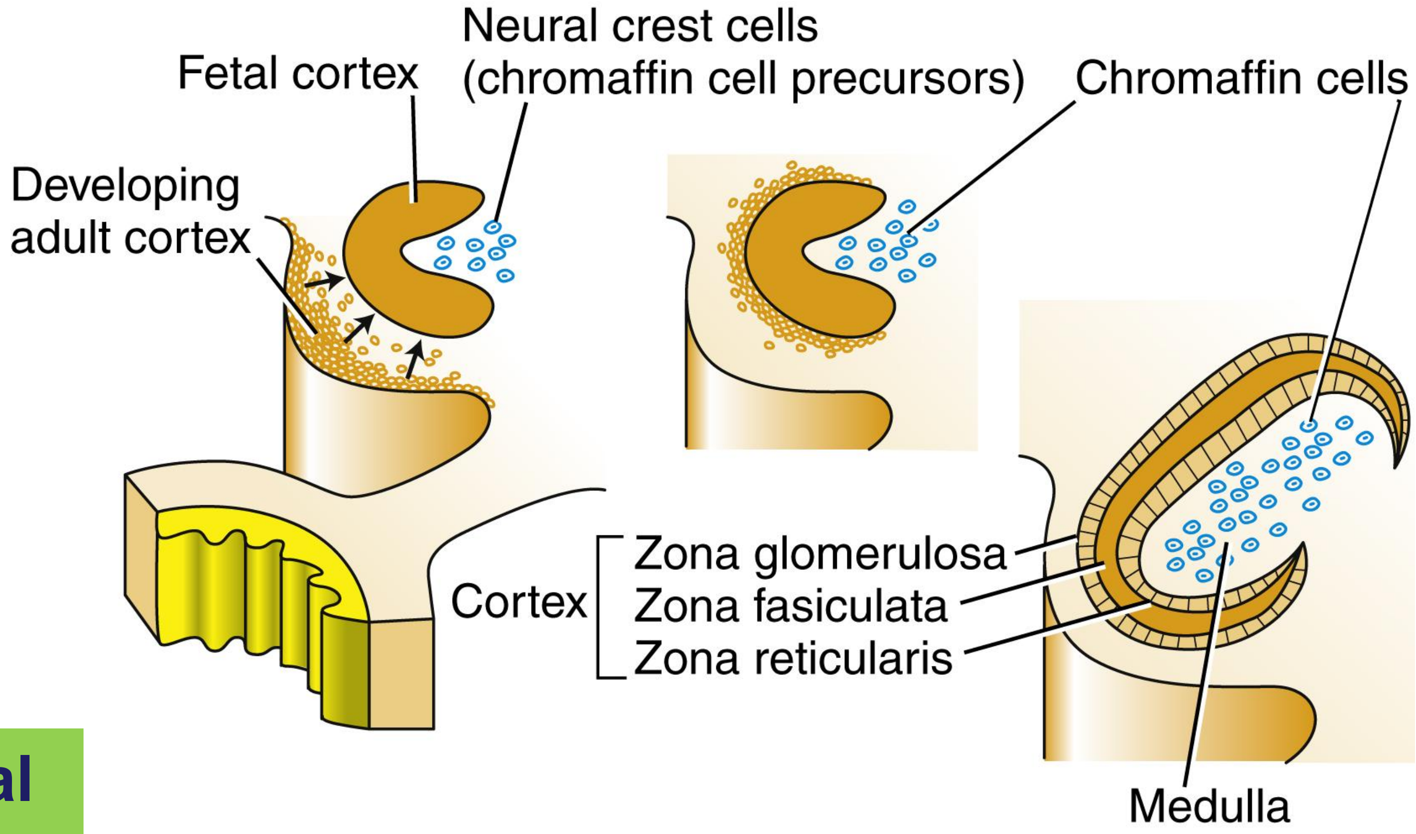
Development of the suprarenal glands

- The suprarenal gland is fully encapsulated by week 8
- **Second trimester** = fetal cortical layer grows rapidly
    - Begins to secrete **dehydroepiandosterone (DHEA)** converted to estrogen by the placenta (helps to maintain pregnancy)
    - Other products **influence the development of the lungs, liver and digestive tract**
- As development continues the gland decreases in size gradually

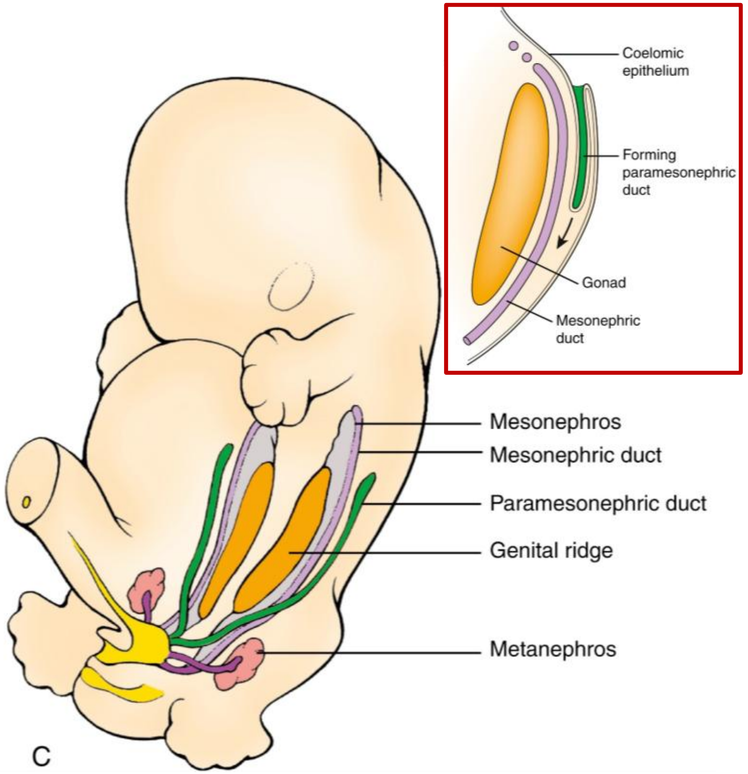


Relative to body mass the fetal suprarenal glands are 10-20x larger than the adult

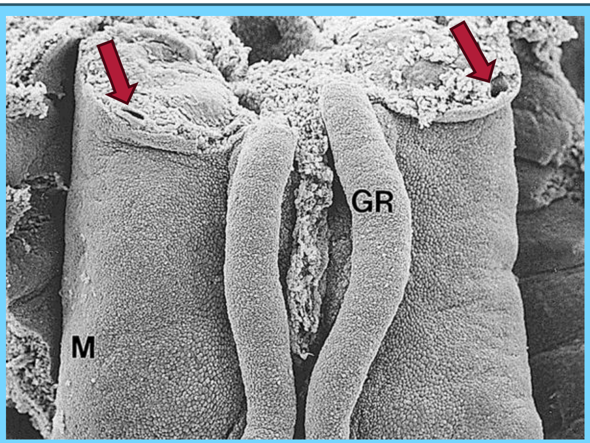
Week 3-6 - Ambiguous/bipotential phase of development

Genital ducts

- **Mesonephros and mesonephric duct** – develops within the intermediate mesoderm in response to the degeneration of the pronephros (week 4)
- **Paramesonephric ducts** – proliferation as invaginations of coelomic epithelial cells forming tubes lateral to each mesonephros (week 6)
    - Caudal ends join to form a Y-shaped structure
    - Makes contact with the urogenital sinus causing it to bulge = **sinus tubercle** (week 6)
    - Cranial ends are open serving as a communication with the peritoneal cavity



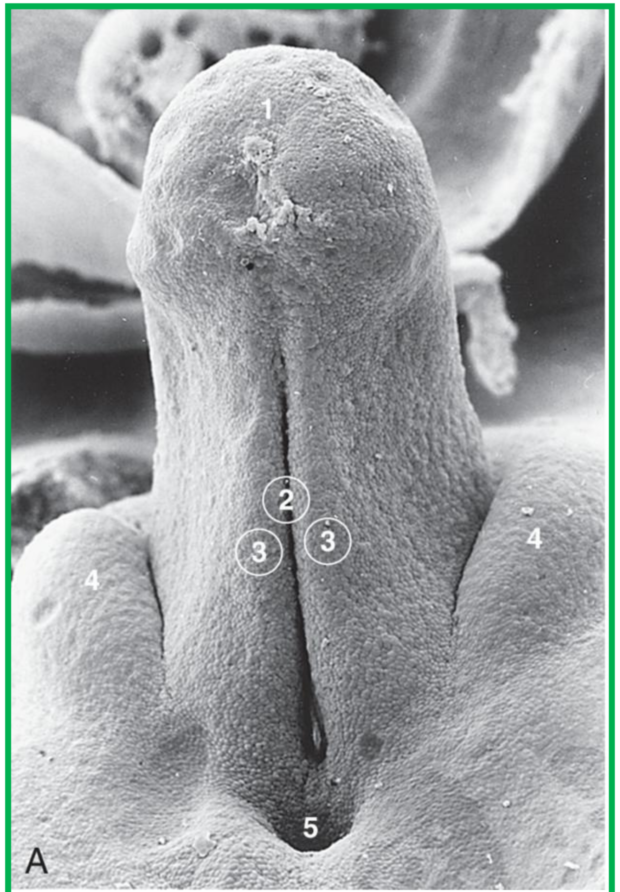
SEM
M = mesonephros
GR = gonadal ridge
mesonephric ducts



Week 3-6 - Ambiguous/bipotential phase of development

External genital organs

- Cloacal folds (mesenchymal swellings) form bilaterally around the cloaca (week 3)
- **Genital tubercle** forms by expansion of the mesoderm anteriorly
- Urorectal septum divides the cloaca into the urogenital sinus and the anorectal canal – completed by week 7
  - Separates the cloacal folds into anterior **urethral folds** and posterior anal folds
- **Labioscrotal swellings** appear bilaterally lateral to the urethral folds



SEM of 7-week embryo

1: Genital tubercle
2: Urethral groove
3: Urethral folds
4: Labioscrotal swellings
5: Anal opening

Development when Y-Chromosome is present

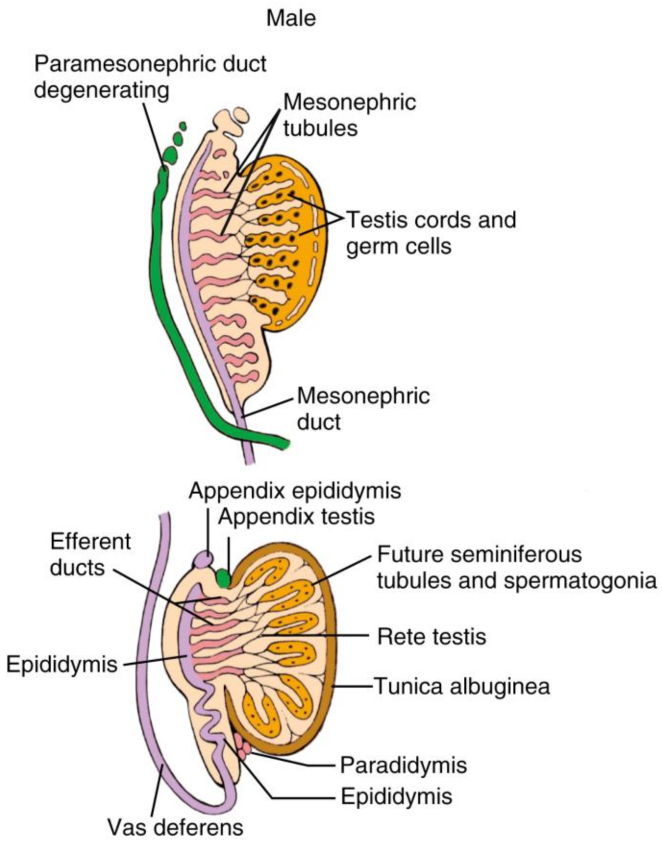
- Testes
- Male genital ducts and accessory male reproductive glands
- Male external genital organs
- Formation of the inguinal canals and testicular descent

Summary of development when Y chromosomes is present

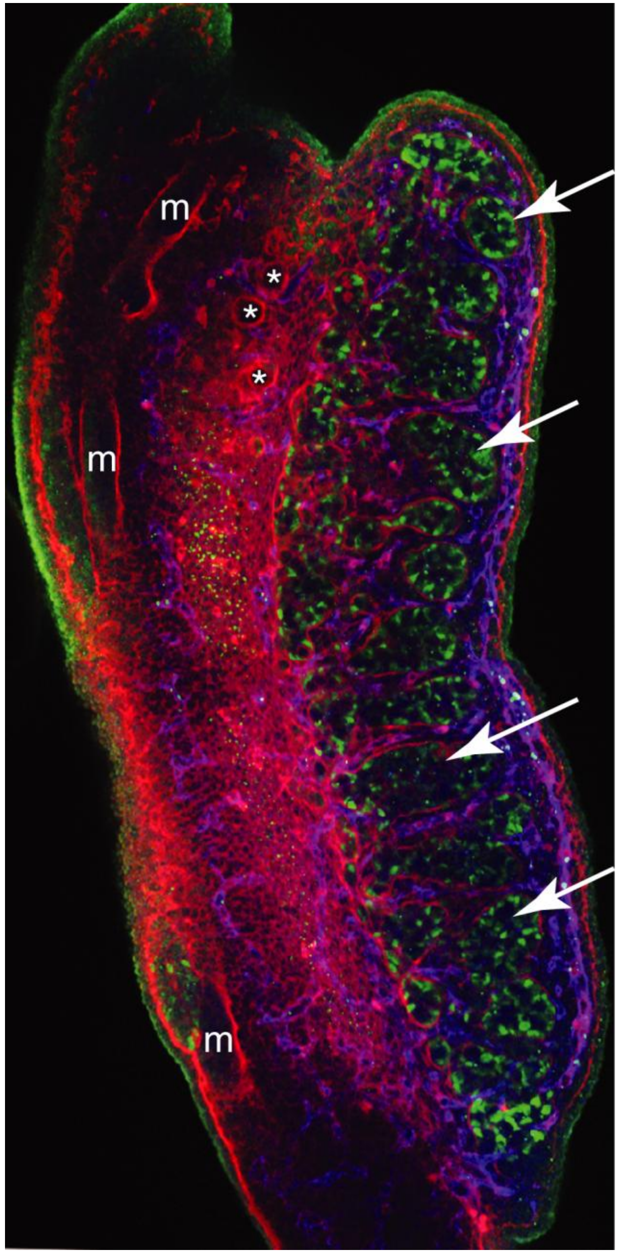
- SRY protein is expressed in the developing gonad (week 6)
- The supporting cells in the gonad differentiate into Sertoli cells and envelope the primordial germ cells
- Sertoli cells, germ cells and myoepithelial cells organize into testes cords which later become the seminiferous ducts
- Sertoli cells in the area closest to the mesonephric duct form series of thin-walled ducts the rete testes
- Rete testes connect with the remaining mesonephric tubules causing a bridge between the seminiferous tubules and ductus deference in the adult
- Sertoli cells produce Anti-Mullerian hormone – causes degeneration of paramesonephric ducts
- Leydig cells develop in the testes, produce Testosterone
- Testosterone is converted to dihydrotestosterone by enzyme 5-Alpha reductase – causes masculinization of the external genital organs
- Mesonephric ducts form the epididymis, ductus deferens and seminal vesicles
- Testes descend into the scrotum because of shortening of the gubernaculum
- Seminiferous tubules and rete testes cannulize at puberty and sperm production begins

Week 7 – in the presence of Y-Chromosome Testes

- Expression of **SRY protein** **by primordial germ cells and pre-Sertoli cells** in developing gonad (Testes determining factor TDF)
- Primitive sex cords proliferate to form **testis/medullary cords**
  - **Testes cords** = Sertoli + myoepithelial + germ cells
- Somatic support cells (from the coelomic epithelium) differentiate into **Sertoli cells**
  - Envelop primordial germ cells
  - Cause them to stop their mitotic action
- **Endothelial cells** migrate from mesonephros into the developing gonad (blood vessel formation)
- **Myoepithelial cells** differentiate from the gonadal ridge mesenchyme
- Sertoli cells form the rete testes
- **Leydig cells** differentiate from the mesenchyme of gonadal ridge
- **Tunica albuginea**, from coelomic epithelium, forms dense connective tissue covering and separates testes from underlying tissue



Developing testes – immunohistochemistry staining

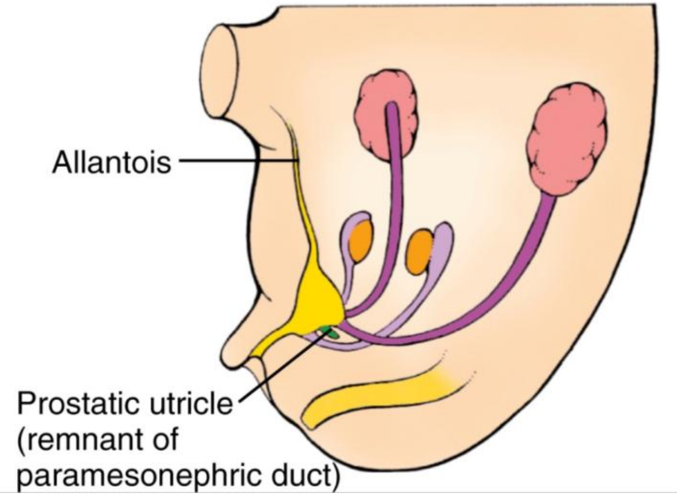


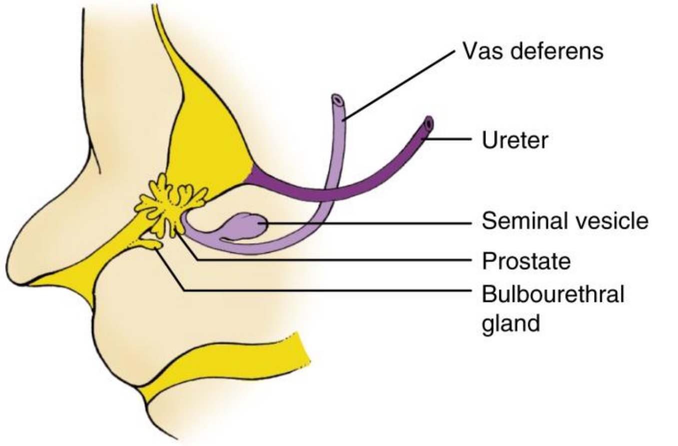
Developing testis cords *(arrows),* demarcated by Sertoli cells (*green*) and outlined by basement membranes *(red)*. Vascularization of the gonad (VE-cadherin, *blue*) Basement membranes also demarcate the mesonephric duct *(m)* and mesonephric tubules *(*)*.

[Larsen's Embryology]

Weeks 8 to 16 – in the presence of Y-Chromosome
Genital ducts

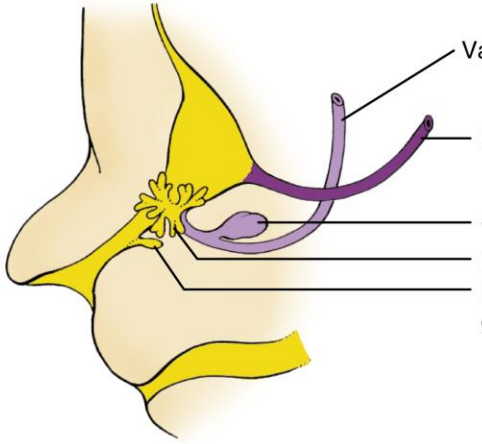
- Sertoli cells secrete **Anti-mullerian hormone** (**AMH**)/mullerian-inhibiting substance (MIS)
  - Causes the rapid regression of paramesonephric ducts
- Leydig cells produce **testosterone** and **androstenedione**
  - Drive the development of mesonephric ductal system
- Mesonephric ducts (md):
  - Cranial portion becomes the **epididymis**
  - 5-12 Mesonephric tubules, adjacent to the epididymis connect with rete testes (complete fusion occurs at 12 weeks) – **efferent ductules**
  - Caudal portion becomes the **ductus deferens**

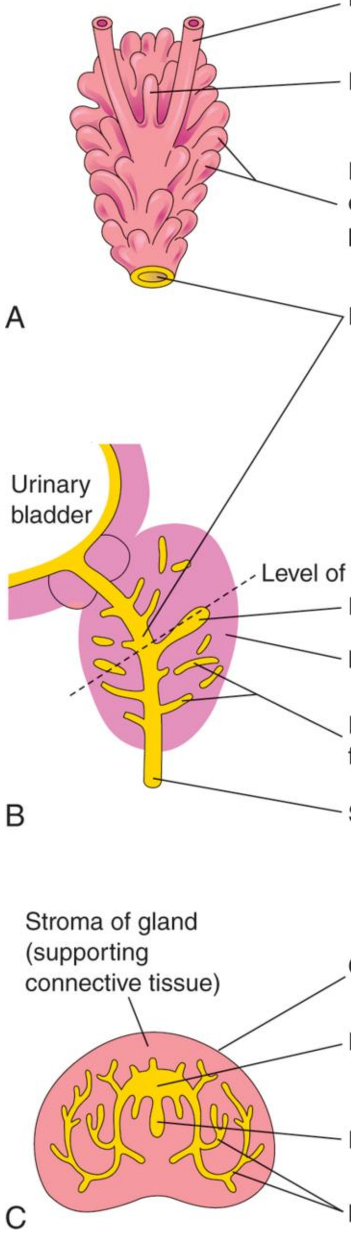




Weeks 8 to 16 – in the presence of Y-Chromosome reproductive glands

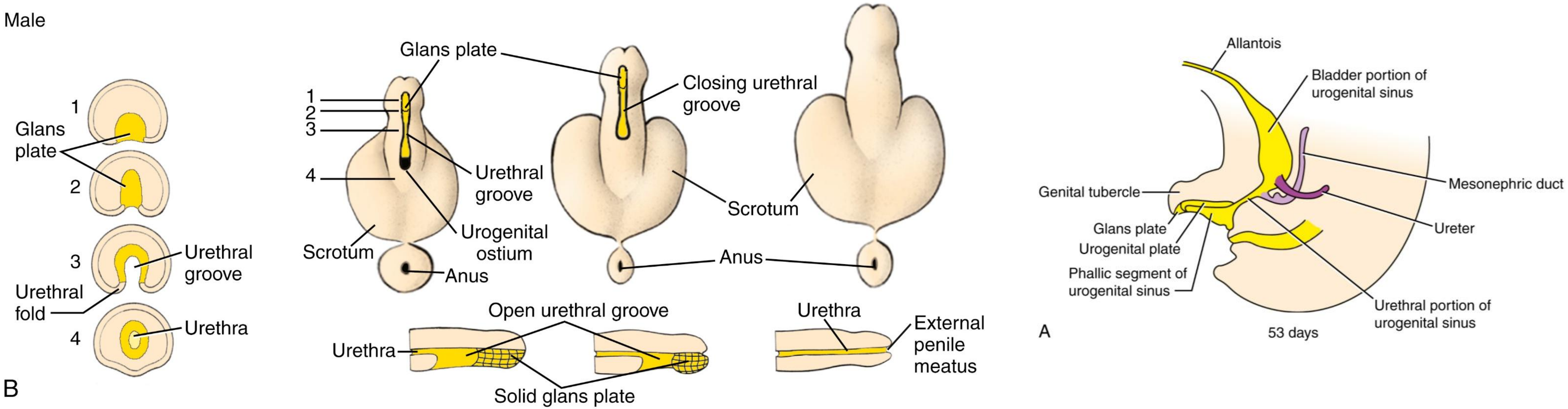
- **Seminal vesicles** grow out from the md close to their junction with urethra (week 10)
  - Ejaculatory duct = portion of md between seminal vesicle and bladder
- **Prostate gland** forms from epithelial outpouchings of the urethra (week 10)
  - Induces changes in the mesenchyme to form smooth muscle and stroma
  - Prostatic utricle is a remnant of the paramesonephric duct
- Testes cords differentiate into **seminiferous cords** (16 weeks)





Weeks 7 to 12 - In the presence of Y-Chromosome External genital organs

- Testosterone is converted to **dihydrotestosterone** by enzyme **5α-reductase** in the Leydig cells
  - Promotes the masculinization of external genital organs
- Penis forms from elongation of primordial phallus
- Urethral plate forms ventrally, cannulizes to form the urethral groove (bounded by the urethral folds)
- Urethral folds and ventral portion of penis fuse (proximal to distal) to form the spongy urethra but does not extend to the tip of the glans



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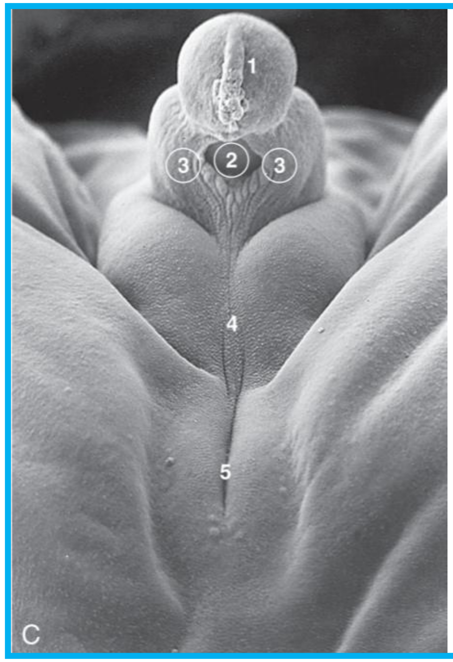
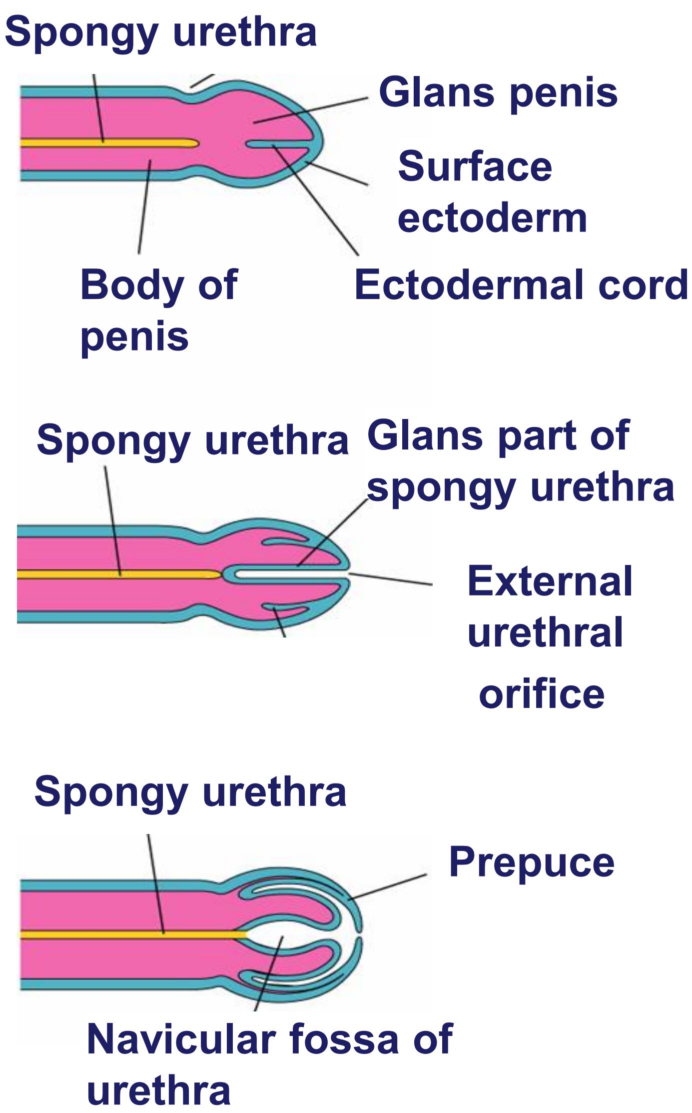
Week 7 onward - In the presence of Y-Chromosome External genital organs cont.

- An **ectodermal cord** forms on the tip of the glans penis grows inwardly to join the spongy urethra
  - The cord canalizes to form the terminal part of the urethra and external urethral orifice
- A circular ectodermal ingrowth at the glans forms the **prepuce** when it breaks down – week 12

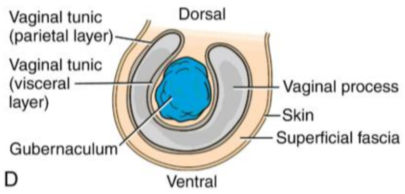


SEM of 10-month male fetus

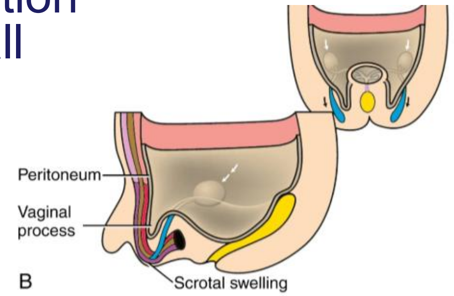
1: Glans penis with ectodermal cord
2: Remains of urethral groove
3: Urethral folds in the process of closing
4: Labioscrotal swellings fusing to form the scrotal raphe
5: Anus

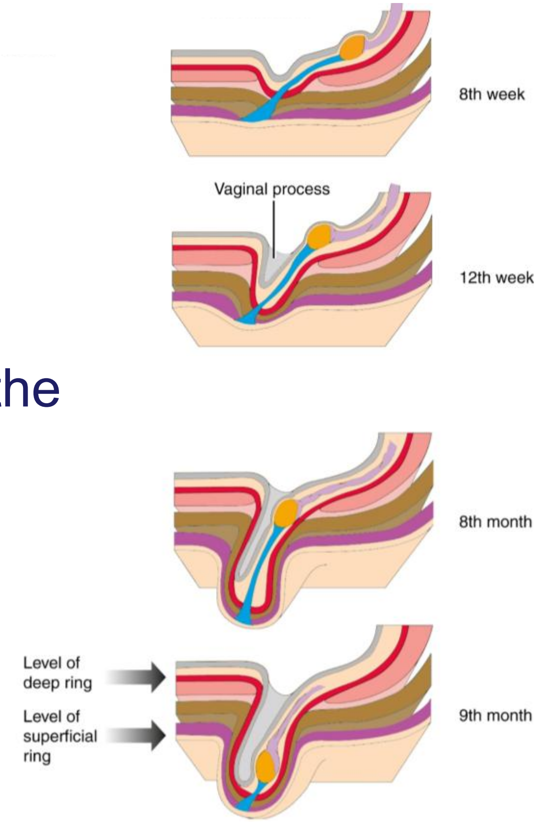


Development of the inguinal canal and testicular descent



- An evagination of the peritoneum, the **processus vaginalis**, develops around the ventral aspect of the gubernaculum
  - It hollows out the inguinal canals and the labioscrotal swellings
  - It grows inferiorly pushing out a sock-like evagination that consist of several layers of the abdominal wall
    - Transversalis fascia
    - Internal oblique muscle and fascia
    - External oblique muscle





- The **gubernaculum** shortens and pulls the testes inferiorly along the abdominal wall, from 3rd to 7th month the testes are situated just cranial to the deep ring

- From the 7th month to the 9th month, the gubernaculum swells (causing a widening of the inguinal canal) and shortens more, guiding the testes through the canal (androgen dependent) and into the scrotum

- The cranial suspensory ligament regresses

Male development at and after birth

- Testes have fully descended into the scrotum (at birth)
- Primordial germ cells differentiate into type A spermatogonia (3 Months)
- The cranial portion of processus vaginalis obliterates and loses its connection to the peritoneal cavity at or just after birth (by end of the 1st year)
- Prepuce gradually detaches from the glans and becomes retractable between the ages of 2years and 4years
- Canulization of the seminiferous tubules, rete testes and efferent ductules (at puberty)

Development when Y-Chromosome is Absent

- Ovaries
- Uterus
- Vaginal canal
- Female external genital organs
- Final positioning of the ovaries and formation of the female ligaments

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Summary of development when Y chromosome is absent

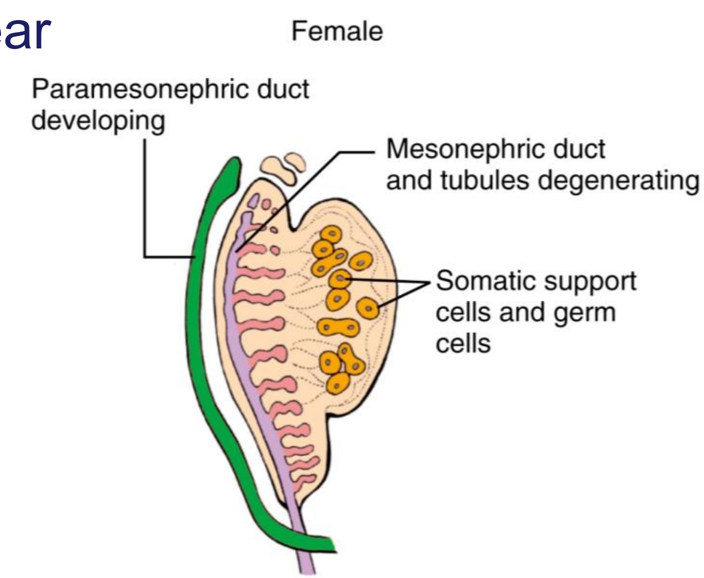
- NO SRY protein is expressed
- Primitive sex cords disassociate
- Primordial germ cells continue to proliferate
- Under the influence of retinol from the mesonephric ducts, primordial germ cells differentiate into oogonia
- Oogonia enter meiosis and become oocytes
- Cortical cords form and dissociate into follicular cells which surround the oocyte preventing further differentiation
- Theca cells develop from the ovary stroma
- Mesonephric duct disappears
- Paramesonephric ducts persist and caudal ends fuse completely to form uterine cavity
- Sinus tubercle expands and becomes a solid mass of endodermal tissue forming the vaginal plate
- The vaginal plate cannulizes and repositions to take its place posterior to the urethra
- External genital organs form by regression of the clitoris and non-fusion of the urethral and labioscrotal folds
- Ovarian descent is linked to the attachment of the gubernaculum to the uterus and repositioning of the vagina

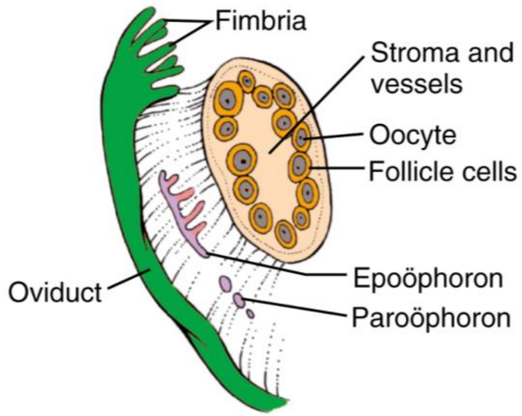
Weeks 7 to 12 – in the **absence** of Y-Chromosome Ovary

- Primitive sex cords dissociate into irregular cell clusters and disappear
- Primordial germ cells proliferate and differentiate into **oogonia** and form clusters
- Oogonia differentiate and enter the 1st meiotic division to become **oocytes**
  - Approx. 2 million at birth

Influenced by retinoic acid from mesonephros

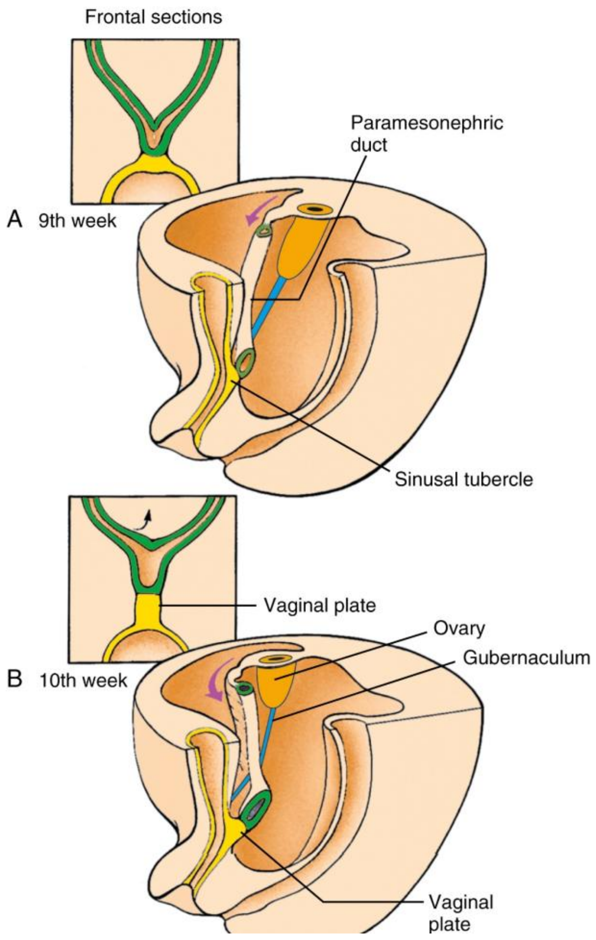
- Surface epithelium proliferates and penetrate the mesenchyme to form **cortical cords**
  - Split into isolated cell clusters which surround the oocyte known as **follicular cells** (week 12)
- **Primordial follicle** = oocyte + surrounding follicular cells
  - Found in the outer cortical region of ovary
  - Follicular cells prevent further differentiation of oocytes
- The stroma between primordial follicles develop into **theca cells**
  - Androgen producing cells





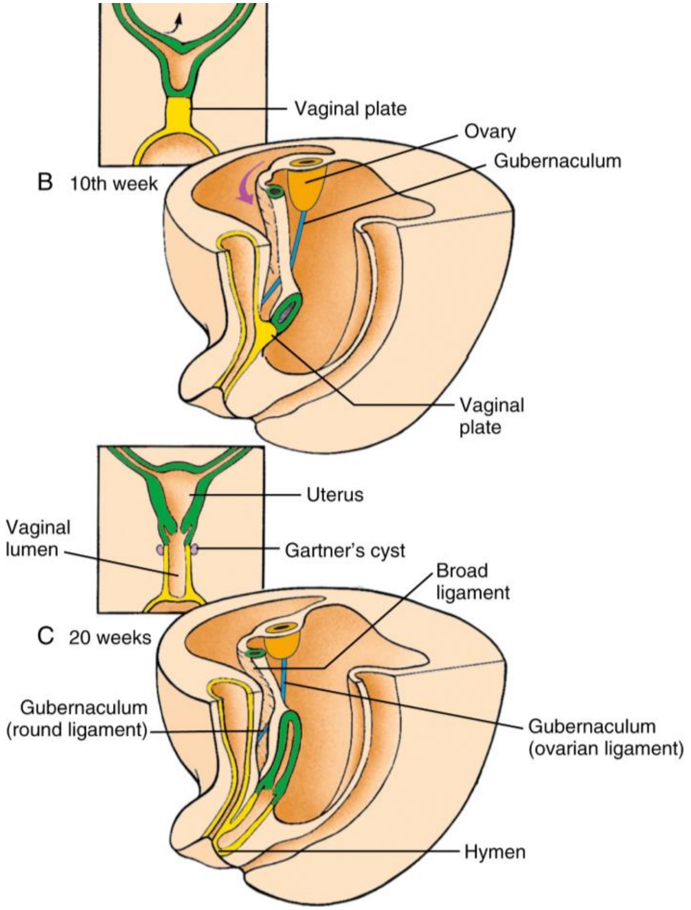
Weeks 9 to 20 – in the absence of Y-Chromosome Genital ducts

- Mesonephric duct disappears
- Y-shaped caudal portion of the paramesonephric ducts fuse completely to form a single cavity, the **uterine cavity** (week 9)
- The remaining unfused portion of the paramesonephric ducts becomes the **uterine tubes**
- Splanchnic mesenchyme give rise to the cells that form the **myometrium** and stroma of the **endometrium**
- The **vaginal plate** forms as the expansion of the sinus tubercle which forms a solid mass of endodermal tissue
  - Lengthening of the vaginal plate pushes the developing uterus further away from the urogenital sinus



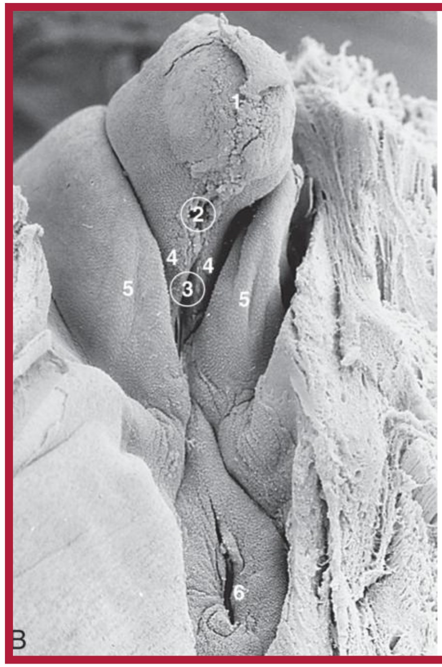
Weeks 9 to 20 – in the absence of Y-Chromosome Genital ducts cont.

- As it elongates the junction between the uterus and vagina causes formation of the **fornix**, which is derived from the paramesonephric ducts
- Canulization (by desquamation) of the vaginal plate results in the formation of the **vaginal lumen/canal**
- The vaginal lumen is separated from the urogenital sinus by an endodermal membrane that will later form the **hymen**
- Further growth and development of the vagina causes translocation resulting in the position posterior to the urogenital sinus and ensuring separate openings of the vagina and urethra (week 20)
  - This also aids the repositioning of the uterine tubes and ovaries



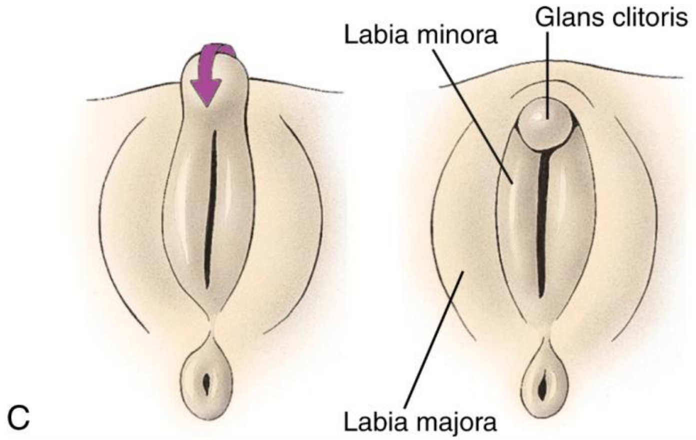
Weeks 7 to 12 - In the absence of Y-Chromosome External genital organs

- Genital tubercle does not lengthen
  - The phallus bends inferiorly to form the clitoris
- Urethral folds remain unfused and become the labia minora
- The labioscrotal folds remain unfused and become the labia majora
- Phallic portion of the urogenital sinus becomes the vestibule of the vagina



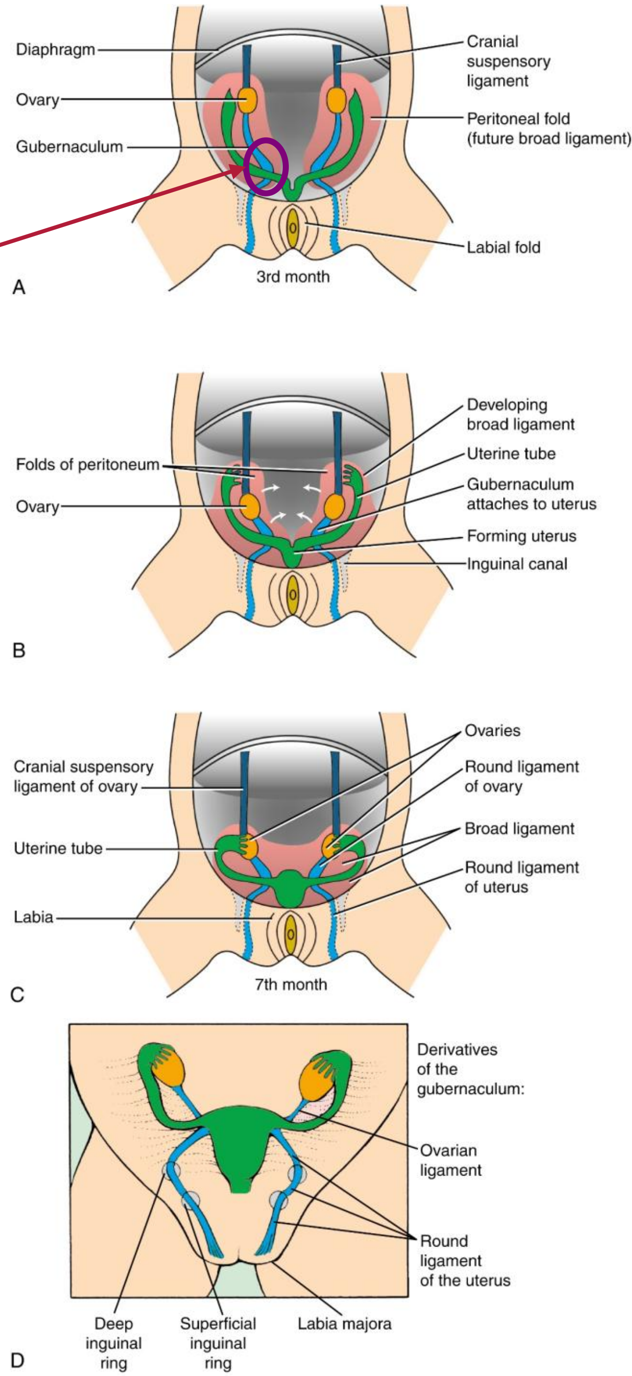
SEM of 10-month female fetus

1: Glans clitoris
2: External urethral orifice
3: Opening into urogenital sinus
4: Urethral folds (labia minora)
5: Labioscrotal swellings (labia majora)
6: Anus



Final positioning of the ovaries and formation of the female ligaments

- The gubernaculum becomes fused to the paramesonephric duct where they cross (week 7)
- As the paramesonephric ducts fuse and enlarge to form the uterus they pull the ovaries down and away from the posterior abdominal wall
- Development and translocation of the vagina causes the peritoneum to drape over the developing uterus and uterine tubes forming the **broad ligament**
  - The ovaries are now located inside the broad ligament
- The gubernaculum now forms the **proper ligament of the ovary** and the **round ligament of the uterus** which passes through the inguinal canal
  - The processus vaginalis obliterates before birth
- The peritoneal layer connecting the ovary to the paramesonephric duct after regression of the mesonephros is called the **Mesovarium**
- The cranial suspensory ligament remains



Female development at and after birth

- Surface epithelium of the ovary flattens into a single layer of cells continuous with mesothelium
  - Positions the ovary to be enveloped in the board ligament on one side only
- Surface epithelium becomes separated from the primordial follicles by a thin layer of dense connective tissue = tunica albuginea
- Hymen perforates leaving a thin membrane covering most of the vaginal opening
- Cervix, endometrium and myometrium complete their development at puberty

Adult Derivatives and Vestigial Remnants of Embryonic Male and Female Reproductive Structures.

We use the term **homologues** for these types of structures = structures that are similar in position, structure and evolutionary origin but nor necessarily in function

Presumptive Rudiments	Male Structure	Female Structure
Indifferent gonad	Testis	Ovary
Primordial germ cell	Spermatogonia	Oogonia
Somatic support cell	Sertoli cells	Follicle cells
Stromal cells	Leydig cells	Thecal cells
Gubernaculum	Gubernaculum testis	Round ligament of ovary Round ligament of uterus
Mesonephric tubules	Efferent ducts of testis Paradidymis	Epoöphoron-paroöphoron
Mesonephric duct	Appendix of epididymis Epididymis Vas deferens Seminal vesicle Ejaculatory duct	Appendix vesiculosa Duct of epoöphoron Gartner's duct
Paramesonephric ducts	Appendix of testis	Uterine tubes Uterus
Urogenital sinus	Prostatic and membranous urethra Prostatic utricle Prostatic gland Bulbourethral glands	Vagina Membranous urethra Urethral/paraurethral glands Greater vestibular glands
Genital tubercle	Glans penis Corpus cavernosa of penis Corpus spongiosum of penis	Glans clitoris Corpus cavernosa of clitoris Bulbospongiosum of vestibule
Urogenital folds and urogenital and glans plate	Penile urethra/ventral part of penis	Labia minora
Labioscrotal folds	Scrotum	Labia majora